

will be proportional to the $\hat{\beta}$ s, the regression coefficients should closely represent the effects of the regressors. If the regressors are highly multicollinear, the $\hat{\beta}$ s may be very poor estimates of the effects of individual regressors.

10.2 COMPUTATIONAL TECHNIQUES FOR VARIABLE SELECTION

We have seen that it is desirable to consider regression models that employ a subset of the candidate regressor variables. To find the subset of variables to use in the final equation, it is natural to consider fitting models with various combinations of the candidate regressors. In this section we will discuss several computational techniques for generating subset regression models and illustrate criteria for evaluation of these models.

10.2.1 All Possible Regressions

This procedure requires that the analyst fit all the regression equations involving one candidate regressor, two candidate regressors, and so on. These equations are evaluated according to some suitable criterion and the “best” regression model selected. If we assume that the intercept term β_0 is included in all equations, then if there are K candidate regressors, there are 2^K total equations to be estimated and examined. For example, if $K = 4$, then there are $2^4 = 16$ possible equations, while if $K = 10$, there are $2^{10} = 1024$ possible regression equations. Clearly the number of equations to be examined increases rapidly as the number of candidate regressors increases. Prior to the development of efficient computer codes, generating all possible regressions was impractical for problems involving more than a few regressors. The availability of high-speed computers has motivated the development of several very efficient algorithms for all possible regressions. We illustrate Minitab and SAS in this chapter. The R function `leaps()` in the `leaps` directory performs an all possible regressions methodology.

Example 10.1 The Hald Cement Data

Hald [1952][†] presents data concerning the heat evolved in calories per gram of cement (y) as a function of the amount of each of four ingredients in the mix: tricalcium aluminate (x_1), tricalcium silicate (x_2), tetracalcium aluminoferrite (x_3), and dicalcium silicate (x_4). The data are shown in Appendix Table B.21. These reflect quite serious problems with multicollinearity. The VIFs are:

x1: 38.496
x2: 254.423
x3: 46.868
x4: 282.513

We will use these data to illustrate the all-possible-regressions approach to variable selection.

[†]These are “classical” data for illustrating the problems inherent in variable selection. For other analysis, see Daniel and Wood [1980], Draper and Smith [1998], and Seber [1977].

TABLE 10.1 Summary of All Possible Regressions for the Hald Cement Data

Number of Regressors in Model	p	Regressors in Model	$SS_{\text{Res}}(p)$	R_p^2	$R_{\text{Adj},p}^2$	$MS_{\text{Res}}(p)$	C_p
None	1	None	2715.7635	0	0	226.3136	442.92
1	2	x_1	1265.6867	0.53395	0.49158	115.0624	202.55
1	2	x_2	906.3363	0.66627	0.63593	82.3942	142.49
1	2	x_3	1939.4005	0.28587	0.22095	176.3092	315.16
1	2	x_4	883.8669	0.67459	0.64495	80.3515	138.73
2	3	x_1x_2	579045	0.97868	0.97441	5.7904	2.68
2	3	x_1x_3	12270721	0.54817	0.45780	122.7073	198.10
2	3	x_1x_4	74.7621	0.97247	0.96697	7.4762	5.50
2	3	x_2x_3	415.4427	0.84703	0.81644	41.5443	62.44
2	3	x_2x_4	868.8801	0.68006	0.61607	86.8880	138.23
2	3	x_3x_4	175.7380	0.93529	0.92235	17.5738	22.37
3	4	$x_1x_2x_3$	48.1106	0.98228	0.97638	5.3456	3.04
3	4	$x_1x_2x_4$	479727	0.98234	0.97645	5.3303	3.02
3	4	$x_1x_3x_4$	50.8361	0.98128	0.97504	5.6485	3.50
3	4	$x_2x_3x_4$	73.8145	0.97282	0.96376	8.2017	7.34
4	5	$x_1x_2x_3x_4$	478636	0.98238	0.97356	5.9829	5.00

TABLE 10.2 Least-Squares Estimates for All Possible Regressions (Hald Cement Data)

Variables in Model	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$
x_1	81.479	1.869			
x_2	57.424		0.789		
x_3	110.203			-1.256	
x_4	117.568				-0.738
x_1x_2	52.577	1.468	0.662		
x_1x_3	72.349	2.312		0.494	
x_1x_4	103.097	1.440			-0.614
x_2x_3	72.075		0.731	-1.008	
x_2x_4	94.160		0.311		-0.457
x_3x_4	131.282			-1.200	-0.724
$x_1x_2x_3$	48.194	1.696	0.657	0.250	
$x_1x_2x_4$	71.648	1.452	0.416		-0.237
$x_2x_3x_4$	203.642		-0.923	-1.448	-1.557
$x_1x_3x_4$	111.684	1.052		-0.410	-0.643
$x_1x_2x_3x_4$	62.405	1.551	0.510	0.102	-0.144

Since there are $K = 4$ candidate regressors, there are $2^4 = 16$ possible regression equations if we always include the intercept β_0 . The results of fitting these 16 equations are displayed in Table 10.1. The R_p^2 , $R_{\text{Adj},p}^2$, $MS_{\text{Res}}(p)$, and C_p statistics are also given in this table.

Table 10.2 displays the least-squares estimates of the regression coefficients. The partial nature of regression coefficients is readily apparent from examination of this

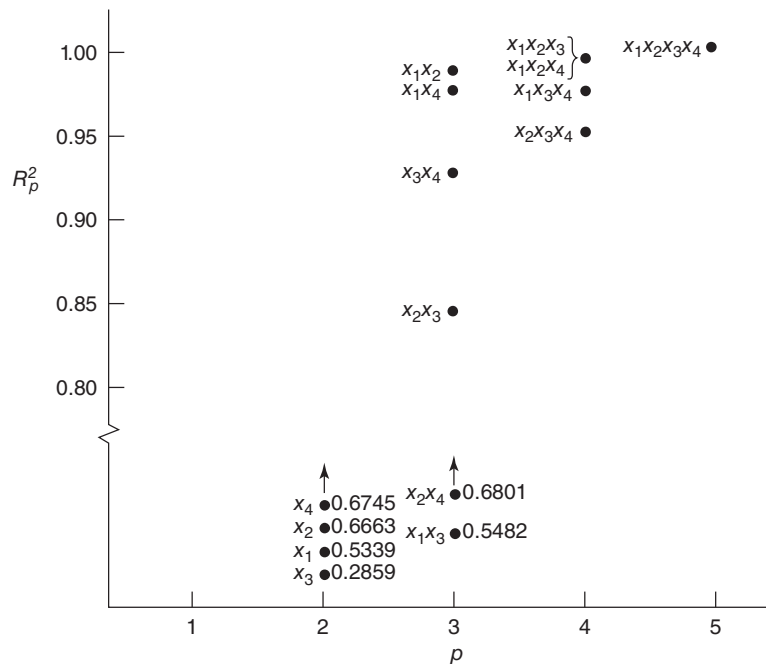


Figure 10.4 Plot of R_p^2 versus p , Example 10.1.

table. For example, consider x_2 . When the model contains only x_2 , the least-squares estimate of the x_2 effect is 0.789. If x_4 is added to the model, the x_2 effect is 0.311, a reduction of over 50%. Further addition of x_3 changes the x_2 effect to -0.923 . Clearly the least-squares estimate of an individual regression coefficient depends heavily on the **other** regressors in the model. The large changes in the regression coefficients observed in the Hald cement data are consistent with a serious problem with multicollinearity.

Consider evaluating the subset models by the R_p^2 criterion. A plot of R_p^2 versus p is shown in Figure 10.4. From examining this display it is clear that after two regressors are in the model, there is little to be gained in terms of R^2 by introducing additional variables. Both of the two-regressor models (x_1, x_2) and (x_1, x_4) have essentially the same R^2 values, and in terms of this criterion, it would make little difference which model is selected as the final regression equation. It may be preferable to use (x_1, x_4) because x_4 provides the best one-regressor model. From Eq. (10.9) we find that if we take $\alpha = 0.05$,

$$\begin{aligned} R_0^2 &= 1 - (1 - R_5^2) \left(1 + \frac{4F_{0.05, 4, 8}}{8} \right) \\ &= 1 - 0.01762 \left[1 + \frac{4(3.84)}{8} \right] = 0.94855 \end{aligned}$$

TABLE 10.3 Matrix of Simple Correlations for Hald's Data in Example 10.1

	x_1	x_2	x_3	x_4	y
x_1	1.0				
x_2	0.229	1.0			
x_3	-0.824	-0.139	1.0		
x_4	-0.245	-0.973	0.030	1.0	
y	0.731	0.816	-0.535	-0.821	1.0

Therefore, any subset regression model for which $R_p^2 > R_0^2 = 0.94855$ is R^2 adequate (0.05); that is, its R^2 is not significantly different from R_{K+1}^2 . Clearly, several models in Table 10.1 satisfy this criterion, and so the choice of the final model is still not clear.

It is instructive to examine the pairwise correlations between x_i and x_j and between x_i and y . These simple correlations are shown in Table 10.3. Note that the pairs of regressors (x_1, x_3) and (x_2, x_4) are highly correlated, since

$$r_{13} = -0.824 \quad \text{and} \quad r_{24} = -0.973$$

Consequently, adding further regressors when x_1 and x_2 or when x_1 and x_4 are already in the model will be of little use since the information content in the excluded regressors is essentially present in the regressors that are in the model. This correlative structure is partially responsible for the large changes in the regression coefficients noted in Table 10.2.

A plot of $MS_{\text{Res}}(p)$ versus p is shown in Figure 10.5. The minimum residual mean square model is (x_1, x_2, x_4) , with $MS_{\text{Res}}(4) = 5.3303$. Note that, as expected, the model that minimizes $MS_{\text{Res}}(p)$ also maximizes the adjusted R^2 . However, two of the other three-regressor models $[(x_1, x_2, x_3)$ and $(x_1, x_3, x_4)]$ and the two-regressor models $[(x_1, x_2)$ and $(x_1, x_4)]$ have comparable values of the residual mean square. If either (x_1, x_2) or (x_1, x_4) is in the model, there is little reduction in residual mean square by adding further regressors. The subset model (x_1, x_2) may be more appropriate than (x_1, x_4) because it has a smaller value of the residual mean square.

A C_p plot is shown in Figure 10.6. To illustrate the calculations, suppose we take $\hat{\sigma} = 5.9829$ (MS_{Res} from the full model) and calculate C_3 for the model (x_1, x_4) . From Eq. (10.15) we find that

$$C_3 = \frac{SS_{\text{Res}}(3)}{\hat{\sigma}^2} - n + 2p = \frac{74.7621}{5.9829} - 13 + 2(3) = 5.50$$

From examination of this plot we find that there are four models that could be acceptable: (x_1, x_2) , (x_1, x_2, x_3) , (x_1, x_2, x_4) , and (x_1, x_3, x_4) . Without considering additional factors such as technical information about the regressors or the costs of data collection, it may be appropriate to choose the simplest model (x_1, x_2) as the final model because it has the smallest C_p .

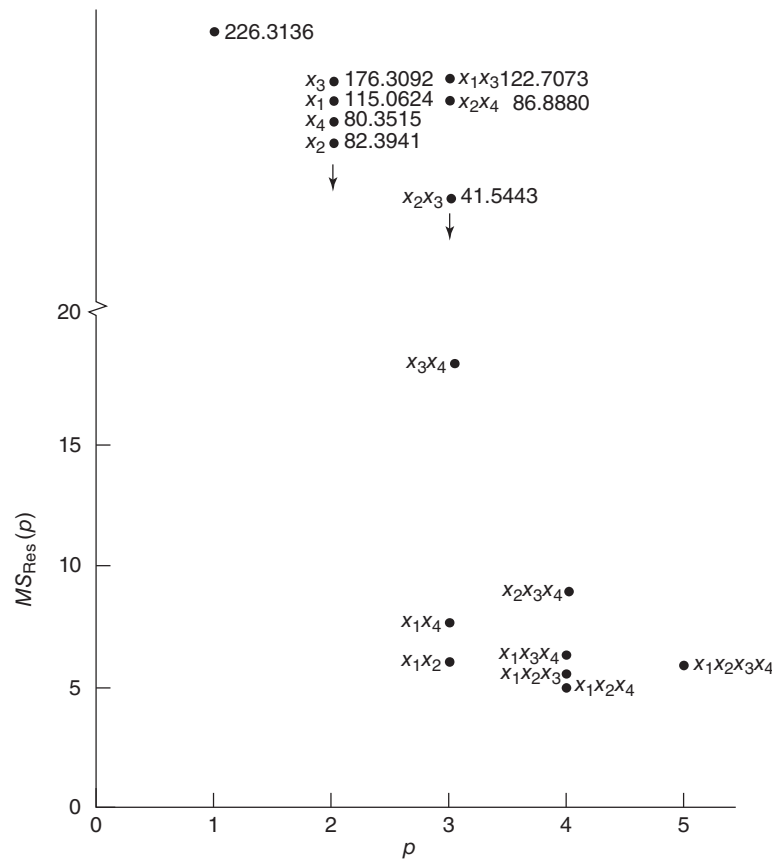


Figure 10.5 Plot of $MS_{Res}(p)$ versus p , Example 10.1.

This example has illustrated the computational procedure associated with model building with all possible regressions. Note that there is no clear-cut choice of the best regression equation. Very often we find that different criteria suggest different equations. For example, the minimum C_p equation is (x_1, x_2) and the minimum MS_{Res} equation is (x_1, x_2, x_4) . All “final” candidate models should be subjected to the usual tests for adequacy, including investigation of leverage points, influence, and multicollinearity. As an illustration, Table 10.4 examines the two models (x_1, x_2) and (x_1, x_2, x_4) with respect to PRESS and their variance inflation factors (VIFs). Both models have very similar values of PRESS (roughly twice the residual sum of squares for the minimum MS_{Res} equation), and the R^2 for prediction computed from PRESS is similar for both models. However, x_2 and x_4 are highly multicollinear, as evidenced by the larger variance inflation factors in (x_1, x_2, x_4) . Since both models have equivalent PRESS statistics, we would recommend the model with (x_1, x_2) based on the lack of multicollinearity in this model. ■

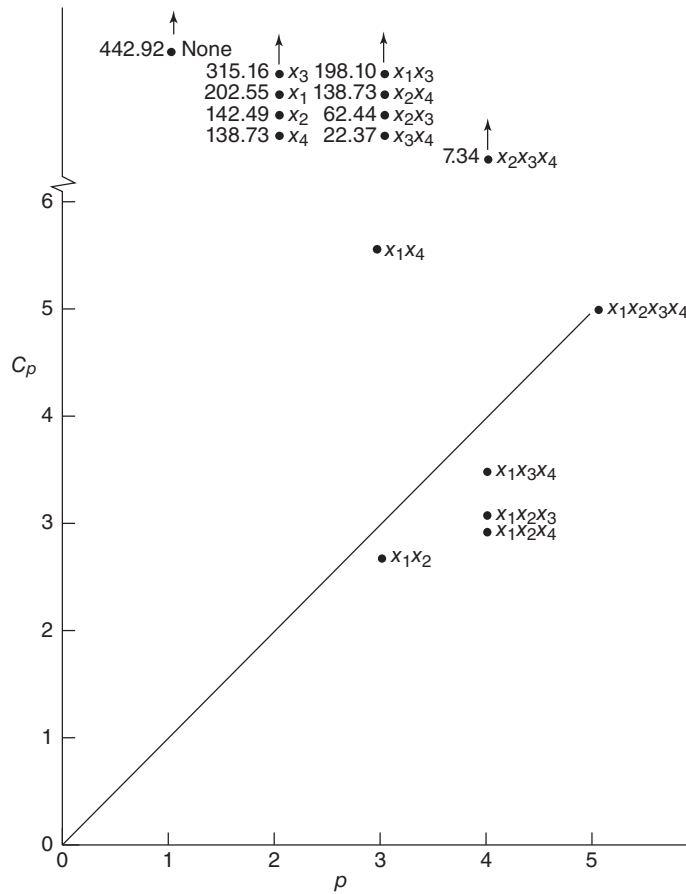


Figure 10.6 The C_p plot for Example 10.1.

Efficient Generation of All Possible Regressions There are several algorithms potentially useful for generating all possible regressions. For example, see Furnival [1971], Furnival and Wilson [1974], Gartside [1965, 1971], Morgan and Tatar [1972], and Schatzoff, Tsao, and Fienberg [1968]. The basic idea underlying all these algorithms is to perform the calculations for the 2^K possible subset models in such a way that sequential subset models differ by only one variable. This allows very efficient numerical methods to be used in performing the calculations. These methods are usually based on either Gauss–Jordan reduction or the sweep operator (see Beaton [1964] or Seber [1977]). Some of these algorithms are available commercially. For example, the Furnival and Wilson [1974] algorithm is an option in many computer programs.

A sample computer output for JMP applied to the Hald cement data is shown in Figure 10.7. This program allows the user to select the best subset regression model

TABLE 10.4 Comparisons of Two Models for Hald's Cement Data

Observation i	$\hat{y} = 52.58 + 1.468x_1 + 0.662x_2^a$			$\hat{y} = 71.65 + 1.452x_1 + 0.416x_2 - 0.237x_4^b$		
	e_i	h_{ii}	$[e_i/(1 - h_{ii})]^2$	e_i	h_{ii}	$[e_i/(1 - h_{ii})]^2$
1	-1.5740	0.25119	4.4184	0.0617	0.52058	0.0166
2	-1.0491	0.26189	2.0202	1.4327	0.27670	3.9235
3	-1.5147	0.11890	2.9553	-1.8910	0.13315	4.7588
4	-1.6585	0.24225	4.7905	-1.8016	0.24431	5.6837
5	-1.3925	0.08362	2.3091	0.2562	0.35733	0.1589
6	4.0475	0.11512	20.9221	3.8982	0.11737	19.5061
7	-1.3031	0.36180	4.1627	-1.4287	0.36341	5.0369
8	-2.0754	0.24119	7.4806	-3.0919	0.34522	22.2977
9	1.8245	0.17195	4.9404	1.2818	0.20881	2.6247
10	1.3625	0.55002	9.1683	0.3539	0.65244	1.0368
11	3.2643	0.18402	16.0037	2.0977	0.32105	9.5458
12	0.8628	0.19666	1.1535	1.0556	0.20040	1.7428
13	-2.8934	0.21420	13.5579	-2.2247	0.25923	9.0194
	PRESS $x_1, x_2 = 93.8827$			PRESS $x_1, x_2, x_4 = 85.3516$		

^a $R^2_{\text{Prediction}} = 0.9654$, $\text{VIF}_1 = 1.05$, $\text{VIF}_2 = 1.06$.

^b $R^2_{\text{Prediction}} = 0.9684$, $\text{VIF}_1 = 1.07$, $\text{VIF}_2 = 18.78$, $\text{VIF}_4 = 18.94$.

of each size for $1 \leq p \leq K + 1$ and displays the C_p , R^2_p , and $MS_{\text{Res}}(p)$ criteria. It also displays the values of the C_p , R^2_p , $R^2_{\text{Adj},p}$ and $S = \sqrt{MS_{\text{Res}}(p)}$ statistics for several (but not all) models for each value of p . The program has the capability of identifying the m best (for $m \leq 5$) subset regression models.

Current all-possible-regression procedures will very efficiently process up to about 30 candidate regressors with computing times that are comparable to the usual stepwise-type regression algorithms discussed in Section 10.2.2. Our experience indicates that problems with 30 or less candidate regressors can usually be solved relatively easily with an all-possible-regressions approach.

10.2.2 Stepwise Regression Methods

Because evaluating all possible regressions can be burdensome computationally, various methods have been developed for evaluating only a small number of subset regression models by either adding or deleting regressors one at a time. These methods are generally referred to as **stepwise-type procedures**. They can be classified into three broad categories: (1) **forward selection**, (2) **backward elimination**, and (3) **stepwise regression**, which is a popular combination of procedures 1 and 2. We now briefly describe and illustrate these procedures.

Forward Selection This procedure begins with the assumption that there are **no regressors in the model** other than the intercept. An effort is made to find

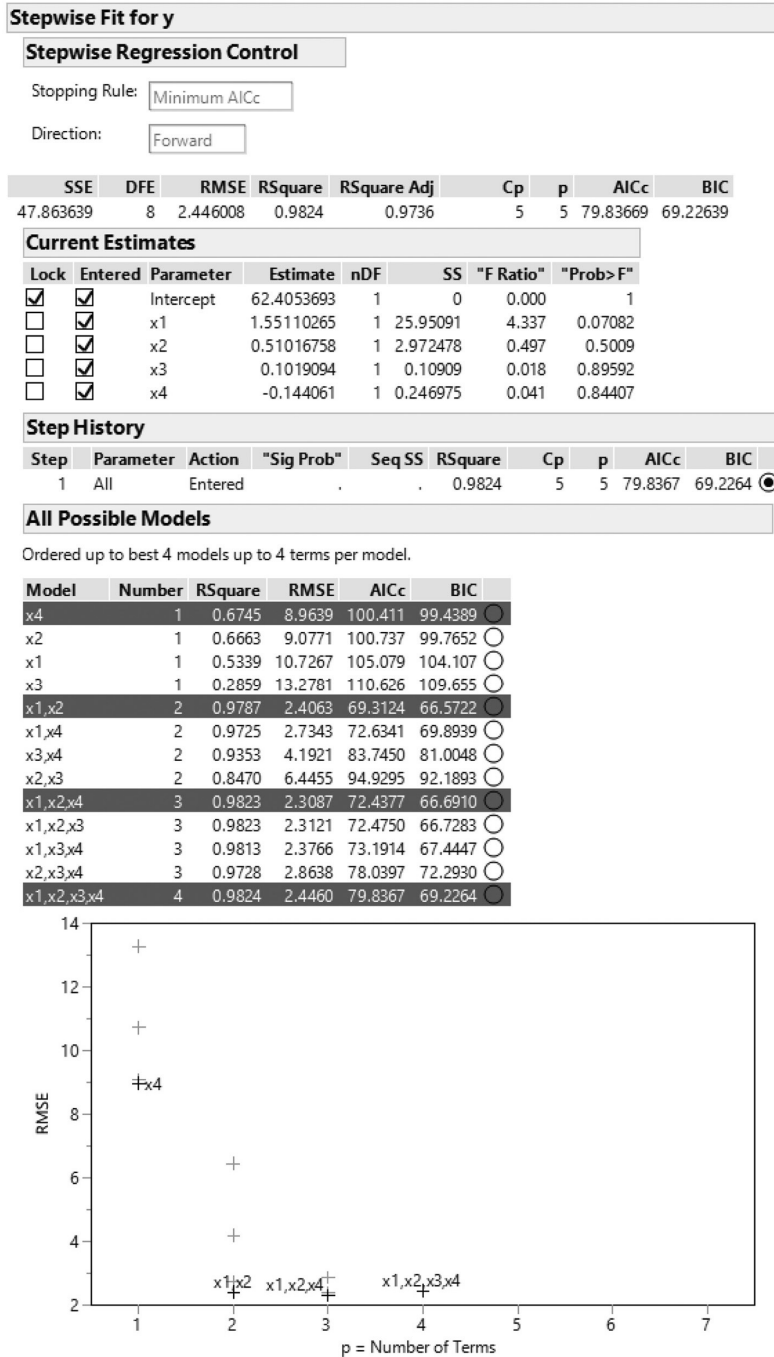


Figure 10.7 JMP computer output for all possible regressions, Hald Cement Data.

an optimal subset by inserting regressors into the model one at a time. The first regressor selected for entry into the equation is the one that has the **largest simple correlation with the response variable y** . Suppose that this regressor is x_1 . This is also the regressor that will produce the largest value of the F statistic for testing significance of regression. This regressor is entered if the F statistic exceeds a preselected F value, say F_{IN} (or F -to-enter). The second regressor chosen for entry is the one that now has the largest correlation with y after adjusting for the effect of the first regressor entered (x_1) on y . We refer to these correlations as **partial correlations**. They are the simple correlations between the residuals from the regression $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$ and the residuals from the regressions of the other candidate regressors on x_1 , say $\hat{x}_j = \hat{\alpha}_{0j} + \hat{\alpha}_{1j} x_1$, $j = 2, 3, \dots, K$.

Suppose that at step 2 the regressor with the highest partial correlation with y is x_2 . This implies that the largest partial F statistic is

$$F = \frac{SS_R(x_2|x_1)}{MS_{\text{Res}}(x_1, x_2)}$$

If this F value exceeds F_{IN} , then x_2 is added to the model. In general, at each step the regressor having the highest partial correlation with y (or equivalently the largest partial F statistic given the other regressors already in the model) is added to the model if its partial F statistic exceeds the preselected entry level F_{IN} . The procedure terminates either when the partial F statistic at a particular step does not exceed F_{IN} or when the last candidate regressor is added to the model.

Some computer packages report t statistics for entering or removing variables. This is a perfectly acceptable variation of the procedures we have described, because $t_{\alpha/2, v}^2 = F_{\alpha, 1, v}$.

We illustrate the stepwise procedure in Minitab. JMP, SAS, and the R function `stepAIC()` in the mass directory also perform this procedure.

Example 10.2 Forward Selection—Hald Cement Data

We will apply the forward selection procedure to the Hald cement data given in Example 10.1. Figure 10.8 shows the results obtained when a particular computer program, the Minitab forward selection algorithm, was applied to these data. In this program the **user specifies the cutoff value for entering variables by choosing a type I error rate α** . Furthermore, Minitab uses the t statistics for decision making regarding variable selection, so the **variable with the largest partial correlation with y is added to the model if $|t| > t_{\alpha/2, v}$** . In this example we will use $\alpha = 0.25$, the default value in Minitab.

Stepwise Regression: y versus x1, x2, x3, x4

Forward selection. Alpha-to-enter: 0.25

Response is y on 4 predictors, with N=13

Step	1	2	3
Constant	117.57	103.10	71.65
x4	-0.738	-0.614	-0.237
T-Value	-4.77	-12.62	-1.37
P-Value	0.001	0.000	0.205
x1		1.44	1.45
T-value		10.40	12.41
P-Value		0.000	0.000
x2			0.42
T-Value			2.24
R-Value			0.052
S	8.96	2.73	2.31
R-Sq	67.45	97.25	98.23
R-Sq (adj)	64.50	96.70	97.64
Mallows C-p	138.7	5.5	3.0

Figure 10.8 Forward selection results from Minitab for the Hald cement data.

From Table 10.3, we see that the regressor most highly correlated with y is x_4 ($r_{4y} = -0.821$), and since the t statistic associated with the model using x_4 is $t = 4.77$ and $t_{0.25/2,11} = 1.21$, x_4 is added to the equation. At step 2 the regressor having the largest partial correlation with y (or the largest t statistic given that x_4 is in the model) is x_1 , and since the partial F statistic for this regressor is $t = 10.40$, which exceeds $t_{0.25/2,10} = 1.22$, x_1 is added to the model. In the third step, x_2 exhibits the highest partial correlation with y . The t statistic associated with this variable is 2.24, which is larger than $t_{0.25/2,9} = 1.23$, and so x_2 is added to the model. At this point the only remaining candidate regressor is x_3 , for which the t statistic does not exceed the cutoff value $t_{0.25/2,8} = 1.24$, so the forward selection procedure terminates with

$$\hat{y} = 71.6483 + 1.4519x_1 + 0.4161x_2 - 0.2365x_4$$

as the final model. ■

Backward Elimination Forward selection begins with **no regressors in the model** and attempts to insert variables until a suitable model is obtained. **Backward elimination** attempts to find a good model by working in the opposite direction. That is, we begin with a model that includes all K candidate regressors. Then the partial F statistic (or equivalently, a t statistic) is computed for each regressor as if it were the last variable to enter the model. The smallest of these partial F (or t) statistics is

compared with a preselected value, F_{OUT} (or t_{OUT}), for example, and if the smallest partial F (or t), value is less than F_{OUT} (or t_{OUT}), that regressor is removed from the model. Now a regression model with $K - 1$ regressors is fit, the partial F (or t) statistics for this new model calculated, and the procedure repeated. The backward elimination algorithm terminates when the smallest partial F (or t) value is not less than the preselected cutoff value F_{OUT} (or t_{OUT}).

Backward elimination is often a very good variable selection procedure. It is particularly favored by analysts who like to see the effect of including all the candidate regressors, just so that nothing “obvious” will be missed.

Example 10.3 Backward Elimination—Hald Cement Data

We will illustrate backward elimination using the Hald cement data from Example 10.1. Figure 10.9 presents the results of using the Minitab version of backward elimination on those data. In this run we have selected the cutoff value by using $\alpha = 0.10$, the default in Minitab. Minitab uses the t statistic for removing variables; thus, a regressor is dropped if the absolute value of its t statistic is less than $t_{\text{OUT}} = t_{0.1/2, n-p}$. Step 1 shows the results of fitting the full model. The smallest t value is 0.14, and it is associated with x_3 . Thus, since $t = 0.14 < t_{\text{OUT}} = t_{0.10/2, 8} = 1.86$, x_3 is removed from the model. At step 2 in Figure 10.9, we see the results of fitting the three-variable

Stepwise Regression: y versus x1, x2, x3, x4

Backward elimination. Alpha-to-Remove: 0.1

Response is y on 4 predictors, with N=13

Step	1	2	3
Constant	62.41	71.65	52.58
x1	1.55	1.45	1.47
T- Value	2.08	12.41	12.10
P- Value	0.071	0.000	0.000
x2	0.510	0.416	0.662
T- Value	0.70	2.24	14.44
P- Value	0.501	0.052	0.000
x3	0.10		
T- Value	0.14		
P- Value	0.896		
x4	-0.14	-0.24	
T- Value	-0.20	-1.37	
P- Value	0.844	0.205	
S	2.45	2.31	2.41
R- Sq	98.24	98.23	97.87
R- Sq (adj)	97.36	97.64	97.44
Mallows C- p	5.0	3.0	2.7

Figure 10.9 Backward selection results from Minitab for the Hald cement data.

model involving (x_1, x_2, x_4) . The smallest $t_{\text{statistic}}$ in this model, $t = -1.37$, is associated with x_4 . Since $|t| = 1.37 < t_{\text{OUT}} = t_{0.20/2,9} = 1.83$, x_4 is removed from the model. At step 3, we see the results of fitting the two-variable model involving (x_1, x_2) . The smallest t statistic in this model is 12.41, associated with x_1 , and since this exceeds $t_{\text{OUT}} = t_{0.10/2,10} = 1.81$, no further regressors can be removed from the model. Therefore, backward elimination terminates, yielding the final model

$$\hat{y} = 52.5773 + 1.4683x_1 + 0.6623x_2$$

Note that this is a different model from that found by forward selection. Furthermore, it is the model tentatively identified as best by the all-possible-regressions procedure. ■

Stepwise Regression The two procedures described above suggest a number of possible combinations. One of the most popular is the **stepwise regression algorithm** of Efroymson [1960]. Stepwise regression is a modification of forward selection in which at each step all regressors entered into the model previously are reassessed via their partial F (or t) statistics. A regressor added at an earlier step may now be redundant because of the relationships between it and regressors now in the equation. If the partial F (or t) statistic for a variable is less than F_{OUT} (or t_{OUT}), that variable is dropped from the model.

Stepwise regression requires two cutoff values, one for entering variables and one for removing them. Some analysts prefer to choose F_{IN} (or $t_{\text{IN}} = F_{\text{OUT}}$ (or t_{OUT}), although this is not necessary. Frequently we choose F_{IN} (or $t_{\text{IN}}) > F_{\text{OUT}}$ (or t_{OUT}), making it relatively more difficult to add a regressor than to delete one.

Example 10.4 Stepwise Regression—Hald Cement Data

Figure 10.10 presents the results of using the Minitab stepwise regression algorithm on the Hald cement data. We have specified the α level for either adding or removing a regressor as 0.15. At step 1, the procedure begins with no variables in the model and tries to add x_4 . Since the t statistic at this step exceeds $t_{\text{IN}} = t_{0.15/2,11} = 1.55$, x_4 is added to the model. At step 2, x_1 is added to the model. If the t statistic value for x_4 is less than $t_{\text{OUT}} = t_{0.15/2,10} = 1.56$, x_4 would be deleted. However, the t value for x_4 at step 2 is -12.62 , so x_4 is retained. In step 3, the stepwise regression algorithm adds x_2 to the model. Then the t statistics for x_1 and x_4 are compared to $t_{\text{OUT}} = t_{0.15/2,9} = 1.57$. Since for x_4 we find a t value of -1.37 , and since $|t| = 1.37$ is less than $t_{\text{OUT}} = 1.57$, x_4 is deleted. Step 4 shows the results of removing x_4 from the model. At this point the only remaining candidate regressor is x_3 , which cannot be added because its t value does not exceed t_{IN} . Therefore, stepwise regression terminates with the model

$$\hat{y} = 52.5773 + 1.4683x_1 + 0.6623x_2$$

This is the same equation identified by the all-possible-regressions and backward elimination procedures. ■

Stepwise Regression: y versus x1, x2, x3, x4

Alpha-to-Enter: 0.15 Alpha-to-Remove: 0.15

Response is y on 4 predictors, with N=13

Step	1	2	3	4
Constant	117.57	103.10	71.65	52.58
x4	-0.738	-0.614	-0.237	
T-Value	-4.77	-12.62	-1.37	
P-Value	0.001	0.000	0.205	
x1		1.44	1.45	1.47
T-Value		10.40	12.41	12.10
P-Value		0.000	0.000	0.000
x2			0.416	0.662
T-Value			2.24	14.44
P-Value			0.052	0.000
S	8.96	2.73	2.31	2.41
R-Sq	67.45	97.25	98.23	97.87
R-Sq(adj)	64.50	96.70	97.64	97.44
Mallows C-p	138.7	5.5	3.0	2.7

Figure 10.10 Stepwise selection results from Minitab for the Hald cement data.

General Comments on Stepwise-Type Procedures The stepwise regression algorithms described above have been criticized on various grounds, the most common being that none of the procedures generally guarantees that the best subset regression model of any size will be identified. Furthermore, since all the stepwise-type procedures terminate with one final equation, inexperienced analysts may conclude that they have found a model that is in some sense optimal. Part of the problem is that it is likely, not that there is one best subset model, but that there are several equally good ones.

The analyst should also keep in mind that the order in which the regressors enter or leave the model does not necessarily imply an order of importance to the regressors. It is not unusual to find that a regressor inserted into the model early in the procedure becomes negligible at a subsequent step. This is evident in the Hald cement data, for which forward selection chooses x_4 as the first regressor to enter. However, when x_2 is added at a subsequent step, x_4 is no longer required because of the high intercorrelation between x_2 and x_4 . This is in fact a **general** problem with the forward selection procedure. Once a regressor has been added, it cannot be removed at a later step.

Note that forward selection, backward elimination, and stepwise regression do not necessarily lead to the **same** choice of final model. The intercorrelation between the regressors affects the order of entry and removal. For example, using the Hald cement data, we found that the regressors selected by each procedure were as follows:

Forward selection	x_1	x_2	x_4
Backward elimination	x_1	x_2	
Stepwise regression	x_1	x_2	

Some users have recommended that all the procedures be applied in the hopes of either seeing some agreement or learning something about the structure of the data that might be overlooked by using only one selection procedure. Furthermore, there is not necessarily any agreement between any of the stepwise-type procedures and all possible regressions. However, Berk [1978] has noted that forward selection tends to agree with all possible regressions for small subset sizes but not for large ones, while backward elimination tends to agree with all possible regressions for large subset sizes but not for small ones.

For these reasons stepwise-type variable selection procedures should be used with caution. Our own preference is for the stepwise regression algorithm followed by backward elimination. The backward elimination algorithm is often less adversely affected by the correlative structure of the regressors than is forward selection (see Mantel [1970]).

Stopping Rules for Stepwise Procedures Choosing the cutoff values F_{IN} (or t_{IN}) and/or F_{OUT} (or t_{OUT}) in stepwise-type procedures can be thought of as specifying a stopping rule for these algorithms. Some computer programs allow the analyst to specify these numbers directly, while others require the choice of a type 1 error rate α to generate the cutoff values. However, because the partial F (or t) value examined at each stage is the maximum of several correlated partial F (or t) variables, thinking of α as a level of significance or type 1 error rate is misleading. Several authors (e.g., Draper, Guttman, and Kanemasa [1971] and Pope and Webster [1972]) have investigated this problem, and little progress has been made toward either finding conditions under which the “advertised” level of significance on the t or F statistic is meaningful or developing the exact distribution of the F (or t)-to-enter and F (or t)-to-remove statistics.

Some users prefer to choose relatively small values of F_{IN} and F_{OUT} (or the equivalent t statistics) so that several additional regressors that would ordinarily be rejected by more conservative F values may be investigated. In the extreme we may choose F_{IN} and F_{OUT} so that all regressors are entered by forward selection or removed by backward elimination revealing one subset model of each size for $p = 2, 3, \dots, K + 1$. These subset models may then be evaluated by criteria such as C_p or MS_{Res} to determine the final model. We do not recommend this extreme strategy because the analyst may think that the subsets so determined are in some sense optimal when it is likely that the best subset model was overlooked. A very popular procedure is to set $F_{\text{IN}} = F_{\text{OUT}} = 4$, as this corresponds roughly to the upper 5% point of the F distribution. Still another possibility is to make several runs using different values for the cutoffs and observe the effect of the choice of criteria on the subsets obtained.

There have been several studies directed toward providing practical guidelines in the choice of stopping rules. Bendel and Afifi [1974] recommend $\alpha = 0.25$ for forward selection. These are the defaults in Minitab. This would typically result in a numerical value of F_{IN} of between 1.3 and 2. Kennedy and Bancroft [1971] also suggest $\alpha = 0.25$ for forward selection and recommend $\alpha = 0.10$ for backward

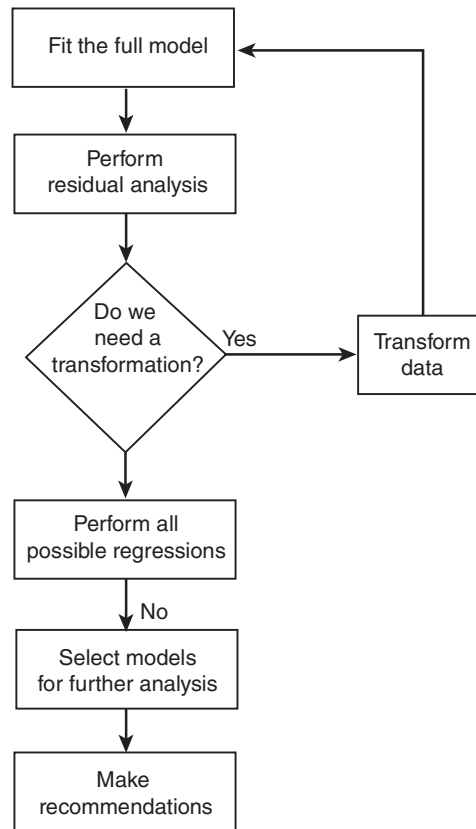


Figure 10.11 Flowchart of the model-building process.

elimination. The choice of values for the cutoffs is largely a matter of the personal preference of the analyst, and considerable latitude is often taken in this area.

10.3 STRATEGY FOR VARIABLE SELECTION AND MODEL BUILDING

Figure 10.11 summarizes a basic approach for variable selection and model building. The basic steps are as follows:

1. Fit the largest model possible to the data.
2. Perform a thorough analysis of this model.
3. Determine if a transformation of the response or of some of the regressors is necessary.
4. Determine if all possible regressions are feasible.
 - If all possible regressions are feasible, perform all possible regressions using such criteria as Mallow's C_p adjusted R^2 , and the PRESS statistic to rank the best subset models.